Experiment 2, Part C: Cell Transfection and Flow Cytometry

Purpose

Test whether each designed binder interacts with its MEGF8 or MOSMO target inside HEK293T cells. The binder is fused to an activation domain and the target to the GAL4 DNA-binding domain. If the binder meets its target, the two halves of the transcription factor are brought together and switch on a GFP reporter, read by flow cytometry. This is a binding screen only: it does not show that a binder disrupts the MMM complex.

Timeline

Day	Step
Before Day 1	Prepare and purify plasmid DNA (Parts A and B)
Day 1	Seed cells
Day 2	Transfect
Day 3	Change media
Day 4	Prepare cells and run flow cytometry

Materials and equipment

- HEK293T cells and HEK293T growth medium
- 96-well tissue culture plates
- Binder plasmids in the activation-domain vector (Parts A and B)
- Target plasmids fused to the GAL4 DNA-binding domain (Parts A and B)
- Reporter plasmid (encodes the GFP reporter and a BFP marker) (Parts A and B)
- GB1-only carrier construct (no target) for the matched negative control
- Echo acoustic liquid handler, 384-well Echo-compatible sterile plates and the Echo transfer spreadsheet
- PEI-MAX (1 µg/µL stock) and sterile 150 mM NaCl
- Plate centrifuge and 37 °C CO_2 incubator
- Access to the university flow cytometry core

Procedure: Day 1, seed cells

1. Seed HEK293T cells at 20,000 cells per well in a 96-well plate with 80 ul per well.
2. Incubate overnight at 37 °C. Cells should be about 70% confluent at transfection.

Procedure: Day 2, DNA mixing and transient transfection

1. Prepare the Echo spreadsheet in advance.
2. Spin the DNA source plate at 500 ×g for 1 minute.
3. Prepare the source stocks and the Echo plate as set out in the spreadsheet.
4. Spin the Echo plate at 1,000 ×g for 1 minute.

5. Combine 150 ng of each plasmid (binder, target and reporter) per well in the destination plate on the Echo, and add any remaining components by hand as the spreadsheet directs.
6. In a sterile hood, make a 1:10 dilution of PEI-MAX in sterile 150 mM NaCl and mix thoroughly by inversion. Use 4.5 µL of the diluted PEI per well.
7. Add sterile 150 mM NaCl to each well to bring the volume to 25 µL.
8. Add the PEI mixture to the DNA wells and mix by pipetting up and down 30 times.
9. Incubate at room temperature for 12–15 minutes, then spin the plate briefly at 1000 xg for 4 minutes to bring the liquid to the bottom.
10. Add the mixture to the cell plate very slowly, almost dropwise, without touching the liquid in the cell plate. Tilt the plate toward you and hold the pipette away from you.
11. Return the plate to the 37 °C incubator.

Procedure: Day 3, media change

1. Change the media very gently.

Procedure: Day 4, flow cytometry

1. Collect the cells. The cells are not fixed. Resuspend the cells in PBS with 3% FBS and 0.5 mM EDTA, and keep on ice, protected from light, until analysis.
2. Flow cytometry is performed by the university flow cytometry core.
3. Gate on BFP-positive cells, then measure the GFP-positive fraction as a percentage of cells.

Conditions and controls

- Test wells: binder + its target + reporter.
- Matched negative control for each binder: the same binder with the GB1-only carrier construct (no MEGF8 or MOSMO target) + reporter.

Expected readout and analysis

The BFP+/GFP+ (double-positive) percentage of cells for each condition. Binding fold change is the double-positive percentage for a binder–target pair divided by that of its own GB1-only control. The MEGF8 and MOSMO screens are analysed separately. A fold change above one suggests binding; no difference is not evidence that a binder does not bind. Fold change should be read alongside the underlying flow populations and the expression of each fusion protein.